

Analyzing long wearable electrocardiogram recordings: R-spike detection and cycle pattern analysis

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Abstract

Introduction: This paper considers very long ECGs (electrocardiograms), looking mainly at leads with salient QRS complexes (deflections in an ECG that represent electrical activity generated by ventricular depolarization). The study focuses on healthy or asymptomatic individuals in order to identify ECG pattern dynamics of potential interest to clinicians, especially for prevention and in the context of personalized medicine.

Materials and methods: The raw data comes from portable recording devices running for up to twenty-four hours; hence, it tends to be rather noisy, even along segments that might be fairly long. The raw data is intervened minimally: only a linear convolution filter is used or the Fasano–Villani method to correct for the wandering baseline. An automatic procedure is used to detect zones devoid of meaningful ECG signal. In order to detect the R-spikes in the good zones of the ECG, a new methodology is offered, taking advantage of the fact that the amplitude of the R-spike is within a narrow range with respect to the height of the wandering line. The proposed method compares favorably with the Pan–Tompkins algorithm. The cycles of the good zones are grouped in blocks, which are then classified as coherent and non-coherent. Certain averages of cycles comprising coherent blocks are the detected patterns. All computations are in Matlab™ (version 2025a).

Results: Graphics devices are presented that allow for the inspection of large sequences of patterns exhibiting the shape change in the morphology of the cycles as based on their representing patterns.

Conclusions: The paper demonstrates an automatic procedure to detect possible anomalies in portable ECG recordings.

Keywords: ECG, patterns, blocks of cycles, QRS complex

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1. Introduction

Massive ECG recordings are considered, in the range of tens of thousand of cycles that might be fairly noisy. The main goal of this work is to detect and extract patterns that might have value in the context of personalized heart care.

Massive recordings can be gathered with portable devices and smartphone apps and allow for practical follow-up of personal heart activity. In the context of prevention, the detection of ECG patterns, which can be correlated with clinical priors of meaningful conditions, might have prognostic value for early disease detection.

The main drawback is that such recordings are prone to fairly high levels of noise coming from various sources, usually in an unpredictable fashion.

The goal of this paper is to detect patterns in massive noisy ECG recordings, identifying zones where patterns occur. The local zones of the recording where (changes in) patterns show up can then be inspected by human specialists or be fed into numerical algorithms for the identification of special ECG conditions.

Zones of the ECG that display recognizable beat recordings, as opposed to ECG areas that are too contaminated by noise or do not

report electric heart activity, are referred to as good zones. These are then extracted automatically from the ECG recording and are scanned for patterns. The pattern discovery technique is based on the detection of sequences of similar cycles and the location of sites within the recording where they change.

The procedure requires only one lead with a recognizable QRS complexes. This significantly eases the data collection and the processing burden.

The approach stays as close as possible to the originally recorded data. To facilitate pattern detection, the R-spikes must be found. For this the standard procedure is to use the Pan–Tompkins algorithm. Additionally, two alternative procedures are presented, which, besides detecting R-spikes, deal with the wandering baseline problem.

Cycle morphologies of ECGs (i.e., the QRS complex, *P* and *T* waves (the *P* and the *T* waves represent the atrial depolarization and the ventricular repolarization, respectively) and the lengths of various segments and intervals) have been widely studied in connection to various health conditions [1]. The Pan–Tompkins [2] algorithm is the classical computational tool for the detection of QRS complexes. Its Matlab™ implementation [3] is especially useful.

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Previous work in automatic ECG processing is usually based on the comparison with given beat templates that are associated with clinical conditions. For recent reviews of these methodologies see [4, 5]. In spite of automatization efforts, even today, it is not unusual for manual ECG inspection to be employed in many clinical instances. More recently, techniques such as Dynamic Time Warping have also been applied for ECG analysis in special instances [6, 7]. Recently [8, 9], data time warping has also been used for ECG analysis.

ECG characteristics have also been studied extensively in the context of risk assessment for new pharmaceutical drugs [10–12]. This area is of recent special interest due to the development of new regulatory protocols for drug safety [13].

The use of portable ECG recording devices has been considered since 2007 [14] at least, and the potential social impact of building affordable portable ECG technology, particularly for lower-income countries, has been discussed recently [15, 16].

There has also been increased interest in analyzing massive ECG recordings via portable, battery-operated sensors as a promising approach for unveiling early indicators of cardiovascular diseases [17]. Consequently, recent efforts have focused on the development of reliable ECG software suitable for mobile phones [18].

The necessity for long-term ECG recordings has been compellingly argued by Acharya et al. in their 2007 book *Advances in Cardiac Signal Processing* [19]. The editors underscore this need in their preface, stating that cardiac diseases “may not show up all the time, but would manifest at certain irregular (random) intervals during the day. Therefore the study of ECG pattern and heart rate variability has to be carried out over extended periods of time”. They further discuss various types of noise and list diagnostic features particularly useful when using portable devices. Additional information on noise types and denoising techniques can be found in the literature [20–22]. Furthermore, the automatic correction of common human errors, such as electrode misplacement, has been a recent focus of research [23].

As stated in [17] long ECG recordings can be processed with smart-phones, provided that reliable computational algorithms are available. In particular, these algorithms must exhibit strong noise tolerance, as noise is common in large-scale recordings obtained with portable devices. For example, an early paper that investigates noise detection and cancellation in ECG signals collected using such devices is [24]. Also Orphanidou et al. [25] address the automatic assessment of signal quality in portable ECG recordings.

A concise and accessible overview of common noise types in ECGs is presented in [26]. The denoising of ECG signals has been extensively studied [27]. More recent contributions, such as [21, 22], provide comprehensive surveys of contemporary ECG denoising techniques.

2. Materials and methods

This paper analyzes datasets typically recorded using portable devices. The data might be contaminated by the noise of several types, have a wandering baseline—due to motion—contain areas of electric outlet interference and suffer from interruptions due to electrode untightening, as shown in **Figure 1**.

Hence it is not unfrequent to find recording areas, lying within an ECG, where there is no clinically useful signal. These regions are removed from the original recording and the remaining stretches of data are then analyzed. The latter are referred to as “good zones”. The timestamps of the good zones have to be preserved for correlation with patient activity. Each good region is partitioned into blocks, which consist of sequences of heartbeat cycles.

A given block may or may not consist of similar cycles. Blocks that consist mainly of similar cycles will be referred to as coherent blocks. The decision on whether a block does or does not have a pattern depends solely on the cycles of the block and it is determined by some chosen parameters such as a heuristic to manage outliers, standard deviation and norm tolerances, which are computed from the block cycles.

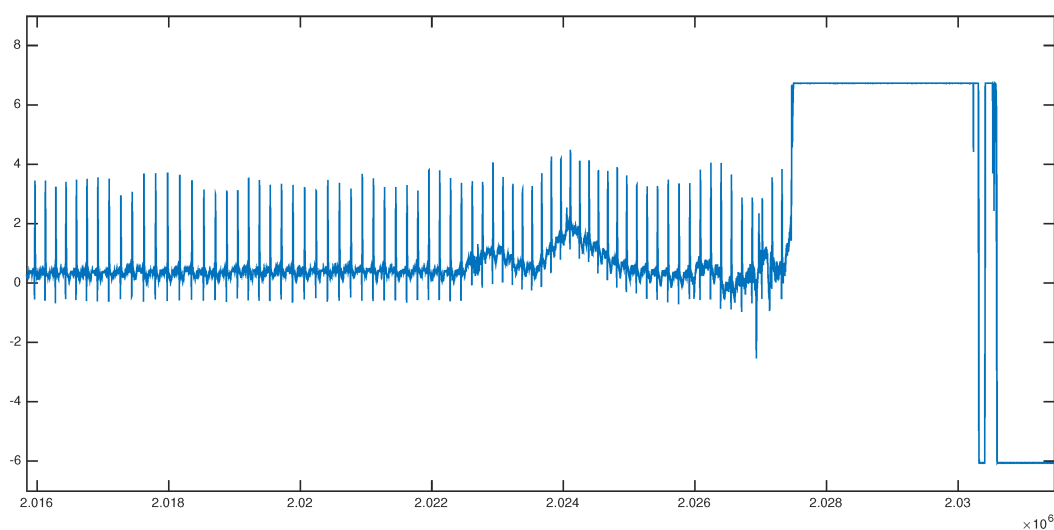


Figure 1 • A problem zone within a massive (approx. 100,000 cycles) ECG; x-axis = numbering of the measurements (sample index, n); y-axis = measurements' intensities (signal amplitude, a.u.).

If a block does have a pattern, this pattern is realized as an average of most of its composing heartbeat cycles. This strategy leads to our goal of detecting patterns by classifying the blocks into those that are coherent and those that are not, within some given tolerance.

Usually an ECG dataset consists of several ECG leads; the peaks of the QRS complex are most easily detectable in the three arm/leg leads. The analysis focuses on any such lead. In the next section we look more closely at the datasets used in this study and describe the methodology.

The first step is to identify the heartbeat cycles within the ECG lead, which can be achieved using the Pan–Tompkins algorithm. Another option is to stabilize the wandering line and then to use a thresholding technique to determine the R-spikes. Two stabilization techniques are examined: a linear convolution filter and the quadratic variability method of Fasano and Villani [28].

- In the case of a linear convolution filter, the kernel depends on the sampling frequency. **Figure 2** displays two examples of kernels and **Figure 3** illustrates the filtering result.

- The quadratic variability method computes the curve representing the wandering isoline and subtracts it from the raw ECG signal. **Figure 4** compares a portion of the original lead with its processed version using the Fasano–Villani method and linear convolution.

An advantage of the second method over the first is that it tends to preserve the shapes more faithfully. The linear convolution filter tends to produce a stronger enhancement of the R-spikes than the second method.

The datasets correspond to ECG recordings of healthy individuals, or subjects for whom—in principle—there is no suspicion of anomalous electric heart behavior. The reason for this is to focus on personal preventive monitoring as opposed to clinical confirmation of disease based on statistically significant symptoms. Our datasets include five of the eighteen long-term recordings of the MIT-BIH Normal Sinus Rhythm Database [29] (the MIT-BIH recordings contain ten million sampling points with a sampling frequency of 128.). See [30] for a recent review that compares various ECG detection and heartbeat classification algorithms that mainly use the MIT-BIH database. Another publicly available source of ECG recordings is [31].

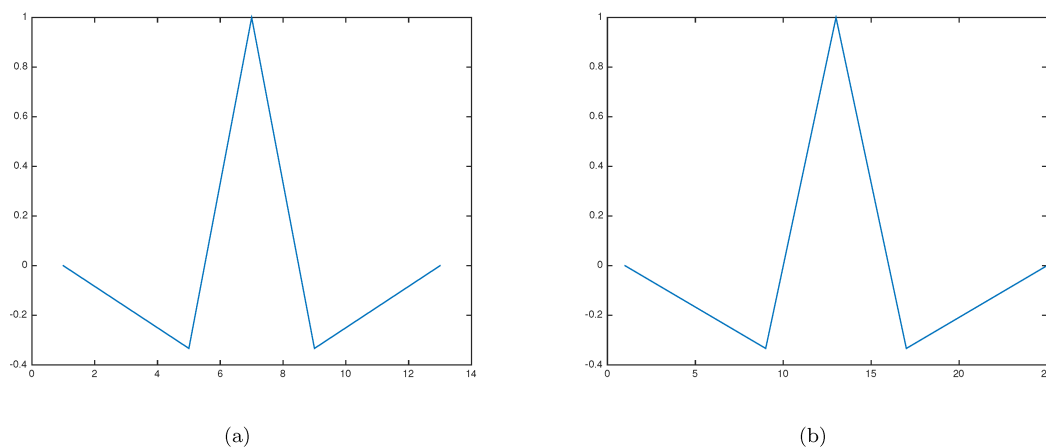


Figure 2 • (a) Kernel for 128 and (b) 250 sample rates; x-axis = numbering of the measurements (sample index, n); y-axis = standard filter scale (filtered signal amplitude, a.u.).

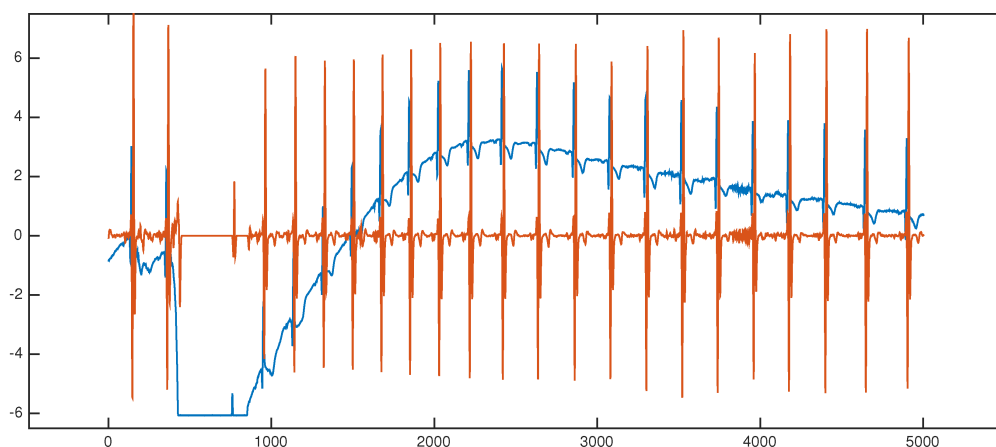


Figure 3 • Raw recording and linear convolution filtered signal; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.).

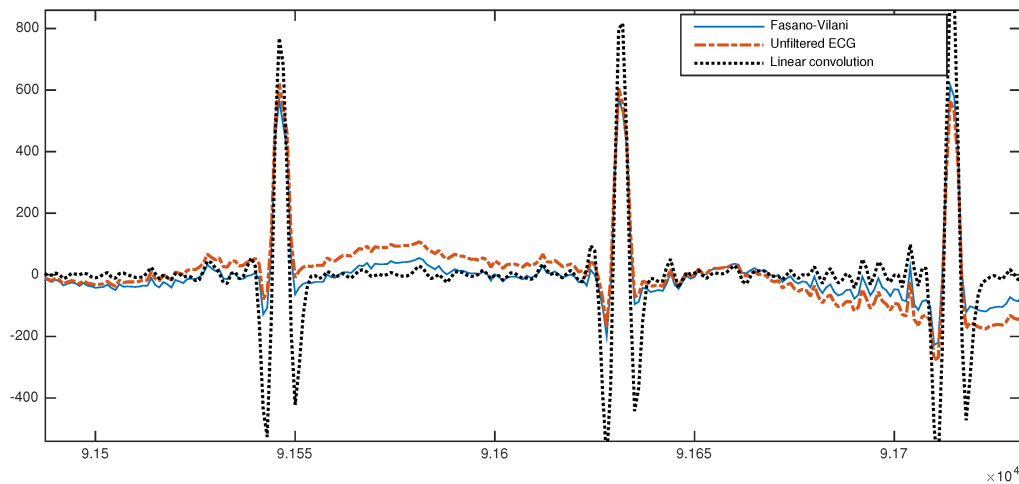


Figure 4 • Effect of Fasano–Villani and linear convolution on the ECG signal; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.), divided by 100, as compared with **Figures 1, 2 and 3**.

The goal of the project is to propose a simple technique especially dedicated to the analysis of noisy and massive ECG recordings, which could lead to the development of a prognosis tool. It will be based on the detection of patterns or irregularities that might have clinical significance. The main effort is directed towards the detection of these patterns, usually using only one arm/leg lead to reduce the processing time. Pattern analysis of other leads is desirable, especially for cross validation, and this will be addressed in future work. As opposed to QRS detection, which is largely based on the Pan–Tompkins algorithm [2], the interest lies in the capture of possibly new, emerging patterns in massive ECGs which are not necessarily connected with any suspected pathology, but still might be interesting to be aware of as a monitoring tool in the context of personalized heart care.

The method consists of the following steps:

- Choose a lead that allows for a robust identification of the QRS complex, usually one of the arm/leg leads. The lead is processed with a linear convolution filter or the Fasano–Villani quadratic variability method to remove the wandering baseline. **Figure 4** illustrates a segment of the 10-million sample 16265 dataset of the MIT-BIH [32] Sinus Normal database that exhibits the unfiltered ECG and the resulting signal after the linear convolution filter and the Fasano–Villani process.
- Estimate a threshold for the detection of R-spike candidates. This is based on the following assumption: the heights of the R-spikes of the filtered lead will tend to cluster within a narrow range, and their number may be roughly estimated. Since it is easy to count the total number of peaks (by peak we mean a local maximum in the ECG looked at as a mathematical graph, not to be confused with an R-spike) greater than any given number λ , the monotone decreasing function $f(\lambda)$, which counts the number of peaks greater than λ , is considered. Then an estimate for the range of heights for the R-spikes can be retrieved from the behavior of f . It corresponds to a plateau in the graph of f , which determines a threshold. **Figure 5** depicts an example of the function f for a recording that consists of about 90 million samples. The sampling rate is 1000 and belongs to a young healthy individual.

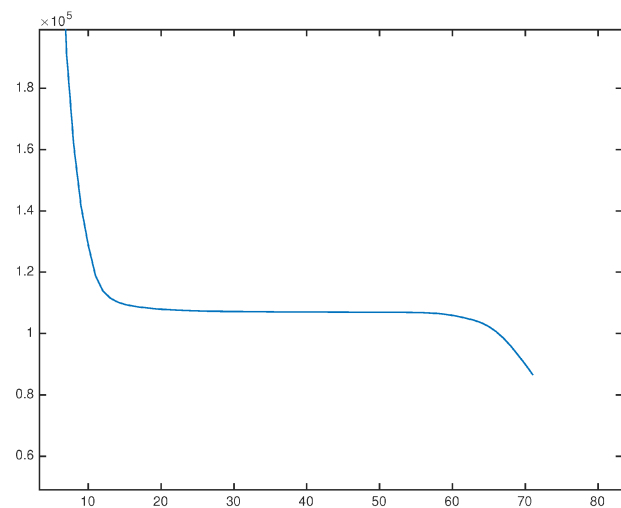


Figure 5 • The monotone function f in terms of a discrete parameter proportional to peak height; x-axis = peak height threshold (λ , signal amplitude, proportional to a.u.); y-axis = number of peaks above the threshold (count, unitless).

- The R-spikes are chosen from the family of peaks that are greater than the threshold satisfying the condition that any two that are contiguous are not too near (a working limit might be one third of the sampling frequency). Experiments suggest that the R-spike computation CPU time for Pan–Tompkins is significantly longer than the above technique and generally the sets of detected spikes are almost exactly the same. A conspicuous exception is the 16272 of the MIT-BIH Sinus Normal database. It consists of 650,000 samples and it is fairly noisy. **Figure 6** exhibits all the R-spikes as detected by both algorithms. The thicker area is a very noisy stretch of the ECG. **Figure 7** shows a transition between a noisy and a not noisy area. **Figure 8** shows good coincidence in an area without noise.
- The goal is to detect emerging patterns, i.e., repetitive sequences of ECG cycles. These sequences of cycles will be housed within the full ECG, may change in length and may be flanked by very noisy areas or areas without a meaningful pattern of heart activity, as illustrated in **Figure 9**. **Figures 10 and 11** zoom into the shown boxes.

Hence, as opposed to signal denoising, this paper is focused on the detection of zones within the full ECG signal, which are not useful for clinical analysis of the heart's electric activity. These zones are removed from the full ECG signal and the resulting clean ECG is analyzed to detect emerging patterns.

- The good zones of the processed lead are estimated as follows. A good zone is a stretch of the ECG such that the distance of two consecutive R-spikes is not greater than twice the sampling rate. In fact, we impose the condition that each good zone should consist of at least fifteen heart beats. Not good zones (i.e., stretches of ECG for which R-spikes have not been detected—such zones arise due to noise, interruptions or large separation between neighboring R-spikes) are gaps between consecutive good zones. This leads to the partitioning of the ECG lead into good zones and gaps between them. Typically, the portions of the recording that separate any two good zones are void of heart electric activity or contain very long, very short or widely separated heart cycles.
- Each good zone is subdivided into blocks. For a lead of a typical ECG recording of a few days duration, there might be about ten thousand blocks of one hundred cycles each. Each

block consists of a fixed number of consecutive cycles, and each cycle is centered at the sample point corresponding to the R-spike and extends towards the neighboring R-spikes to the left and right. Its length depends on these three R-spikes. Half the distance to the right and to the left of neighboring R-spikes is chosen, hence the cycles belonging to a block might have different lengths.

Each block may be labeled as coherent or non-coherent according to whether their cycles are or are not similar, respectively. In some cases, for borderline blocks, it may be convenient not to assign them a definitive coherent or non-coherent classification and save them for further, possibly manual inspection.

Coherent blocks define patterns and non-coherent blocks might be discarded or saved for further processing. **Figure 12** illustrates a 100-cycle block. The R-spikes of the cycles have been aligned and are presented in a 3D setting.

- Pattern evolution is sought as we advance (in time) from one block to the next, contiguous one within a given good zone. This process is realized for each good zone.

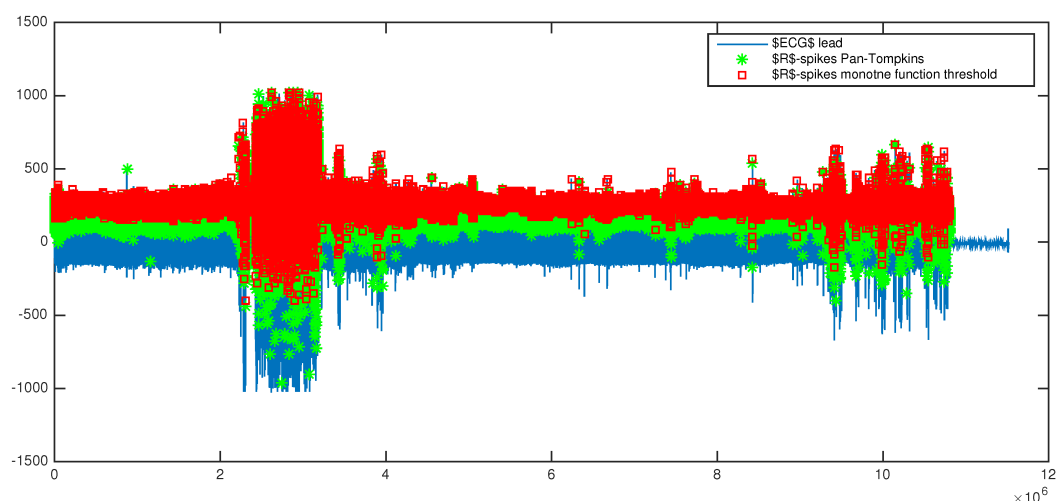


Figure 6 • R-spikes of MIT-BIH Sinus Normal 16272, green 91,317 Pan-Tompkins and 90,289 threshold; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.).

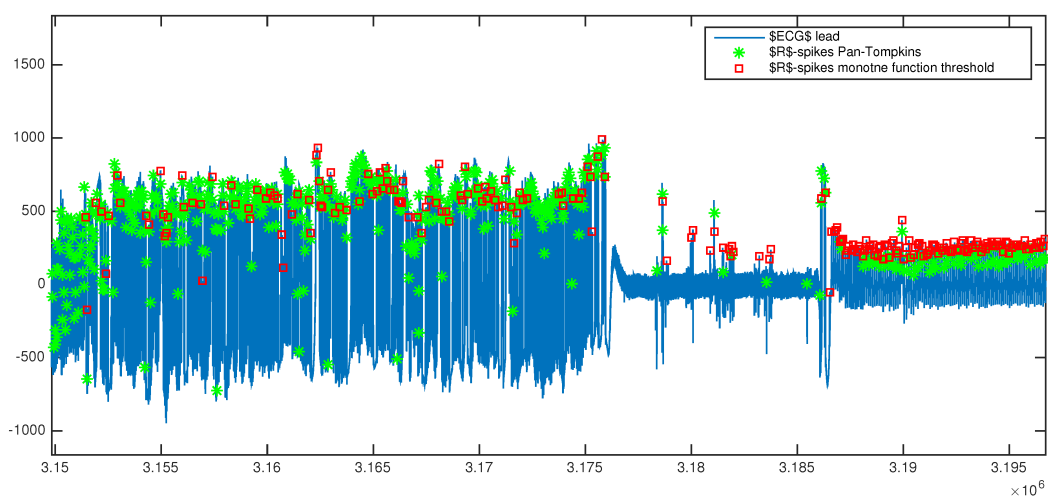


Figure 7 • Close-up of a transition between a noisy and a not noisy area; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.).

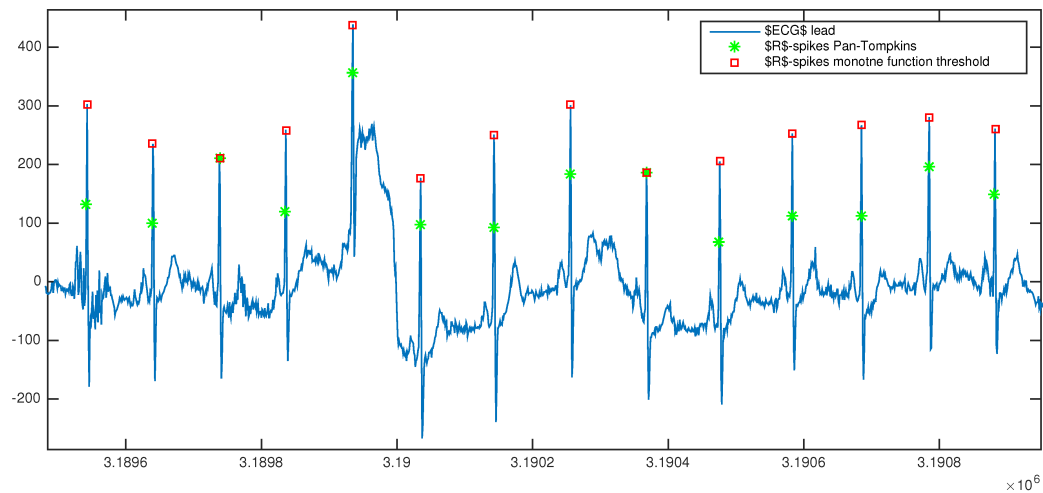


Figure 8 • Coincidence between Pan–Tompkins and threshold; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.).

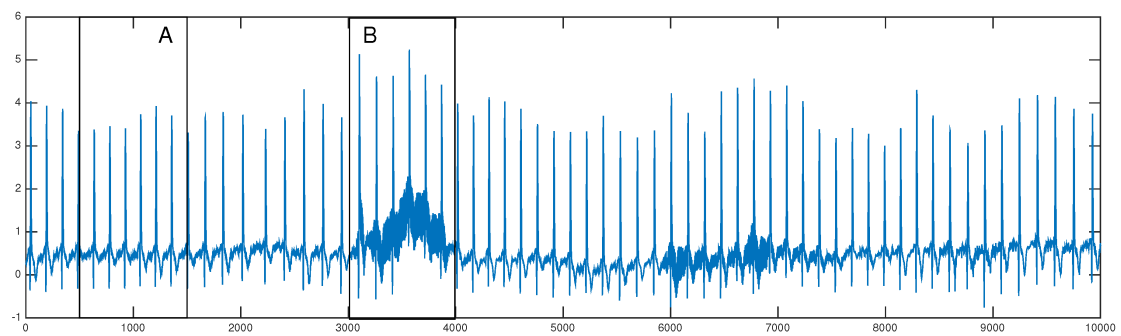


Figure 9 • Pockets of very noisy signal, young healthy adult. Sampling frequency 250 showing 10,000 samples; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.).

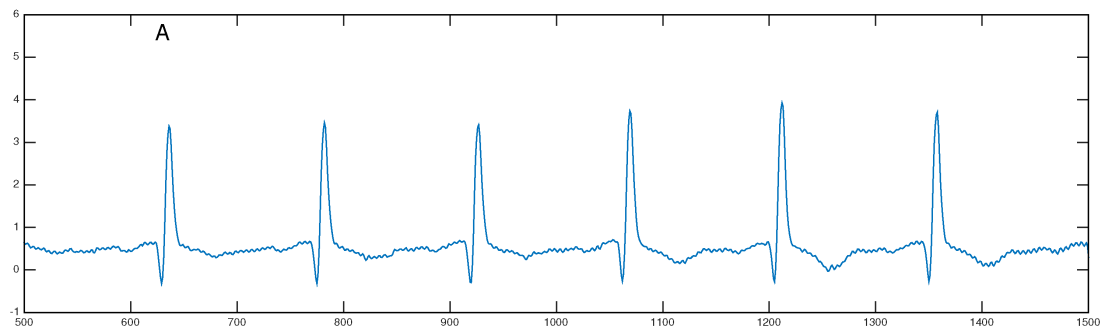


Figure 10 • Zoom of box A of **Figure 9**; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.).

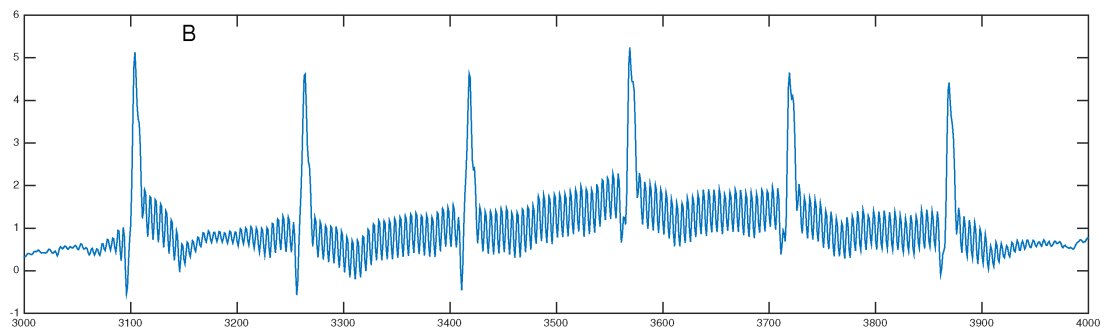


Figure 11 • Zoom of box B of **Figure 9**; x-axis = numbering of the recorded signal (sample index, n); y-axis = recorded height (signal amplitude, a.u.).

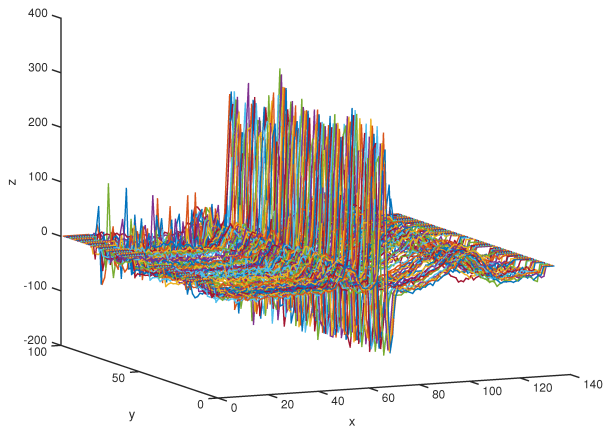


Figure 12 • A 3D view of hundred cycle block; x-axis = signal offset position with respect to the R-spike (sample index relative to R-spike, n); y-axis = depth to differentiate cycles aligned by R-spike (cycle index, unitless); z-axis = corresponding signal height (signal amplitude, a.u.).

The blocks of a good zone are the elements from which patterns arise. Block size is fixed and it usually contains between fifty and two hundred consecutive cycles. The actual choice of the number of cycles per block might be user defined. Smaller block sizes tend to produce larger families of patterns. Larger block sizes tend to smooth out differences between patterns and focus on more global trends in the good zone under inspection.

For each block an average cycle is computed. Namely, for each sample index the average is taken for all cycles within the block that share the index. The cycles of each block are ordered according to their L_∞ distance to the average, taking into account and compensating for the fact that the cycle lengths vary within the block. The top 90% of the block's cycles are used to compute the standard deviation at the sample points. On this is based the decision on whether a block is coherent and hence carries a pattern, is non-coherent and merits further inspection, or is considered

void of useful clinical information. If the block is coherent, its associated pattern is defined to be the average of the top 90% of its cycles. The averages are computed at each sample point taking into account the number of cycles that share the point. The averages are not computed at sample points populated by less than 50% of the cycles of the block.

Block coherence depends on the deviation of the top 90% of the cycles from its average. Namely, the average of the block is increased by its *minimum* + 1, and we compute its quotient by the standard deviation of this increased average for each sample point represented in this average (the ratio of the standard deviation and the average is referred to as the coefficient of variation).

This is done for all samples excluding the points corresponding the QRS complex (the reason for this exclusion is that the error in the data acquisition process tends to increase significantly in the QRS area of the cycles).

The decision on whether a block is coherent or not is based on the maximum of the coefficient of variation (i.e., quotients of the standard deviation and the increased average) at all samples of the average cycle. A threshold for this maximum depends heavily on the noise level and may be user adjusted. If there is no prior knowledge on the noise level, the parameters and tolerances may be adjusted so that about 75% of the blocks living in good zones are classified as coherent.

3. Results

Datasets from the MIT-BIH Normal Sinus Rhythm Database [29] are examined. **Table 1** summarizes the good zones, number of blocks and block coherence for five datasets. Each block consists of one hundred cycles. Block coherence parameters were estimated by visual inspection.

Figure 13 sketches a short stretch of the typical lay out of a good zone presented as sequences of blocks. Each block is composed of 100 cycles.

Table 1 • The numbers in parentheses refer to the total number of samples.

Dataset	Samples in good zones	Sampling rate	QRS	Nr. good zones	Total Nr. blocks	Non coherent blocks
MIT-16483	9,737,628 (11,960,320)	128	104,171	9	1038	205
MIT-16420	9,951,118 (11,051,008)	128	101,891	6	1016	308
MIT-16273	9,516,435 (11,354,112)	128	89,828	2	897	89
MIT-16272	10,357,010 (11,520,000)	128	87,767	91	845	199
MIT-16265	10,247,884 (11,730,944)	128	100,435	1	1004	102

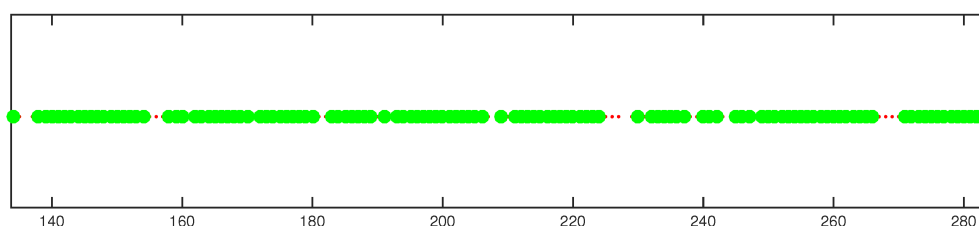


Figure 13 • A typical stretch of the good zone of MIT-16265. It is composed of coherent blocks (heavy green dots), red dots correspond to non-coherent blocks and the absence of dots points to blocks that require further inspection. The dataset contains only one good zone, which consists of 1004 blocks, and each block consists of 100 cycles; x-axis = numbering of blocks (block index, unitless).

Figures 14 and 15 illustrate the most and least coherent blocks among all coherent blocks and **Figures 16 and 17** depict the patterns associated with each of these blocks. **Figure 18** shows a block that does not carry a pattern. The dataset is MIT-16420.

The identified patterns are the averages of cycles of the coherent blocks of the ECG. **Figures 19 and 20** display two sets of chronologically arranged patterns—each pattern is the average of the cycles of its generating block. The blocks consist of 100 and 50 cycles respectively and refer to the same stretch of the MIT-16420 dataset. Both figures convey similar overall pattern presence. Namely, the patterns of the most coherent 100 and 50 cycle blocks indicate the same pattern variation. It may be useful to note that adjacency of the patterns in **Figures 19 and 20** does not imply that they are exactly adjacent in the ECG but it does allow for the visual inspection of the variability of the most numerically faithful (in terms of its shape, or more precisely with the least deviation) of all coherent patterns present in the corresponding segment of the ECG.

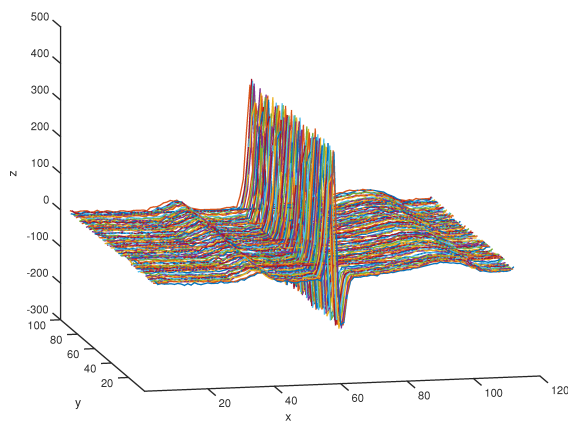


Figure 14 • Most coherent one hundred cycle block of the MIT-16420. The ten worst cycles were replaced by placeholders displayed as straight line segments; x-axis = signal offset position with respect to the R-spike (sample index relative to R-spike, n); y-axis = depth to differentiate cycles aligned by R-spike (cycle index, unitless); z-axis = corresponding signal height (signal amplitude, a.u.).

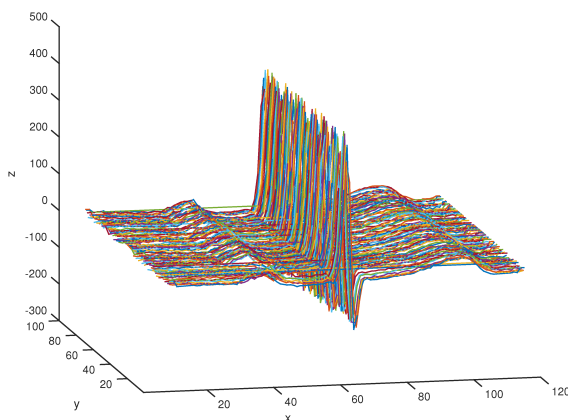


Figure 15 • Least coherent block of all coherent blocks of the MIT-16420 dataset; x-axis = signal offset position with respect to the R-spike (sample index relative to R-spike, n); y-axis = depth to differentiate cycles aligned by R-spike (cycle index, unitless); z-axis = corresponding signal height (signal amplitude, a.u.).

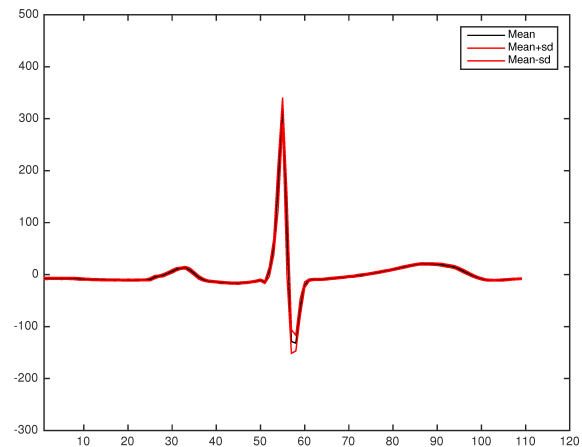


Figure 16 • The pattern associated with the most coherent block of MIT-16420. The standard deviation is minimal; x-axis = numbering of signals of one block (sample index, n); y-axis = signals' heights (signal amplitude, a.u.).

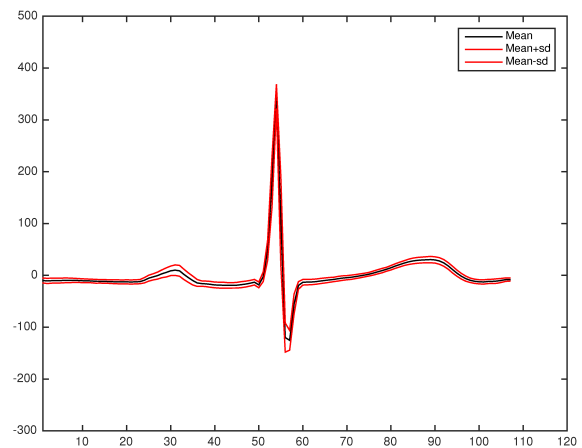


Figure 17 • The pattern associated with the least coherent block of MIT-16420. The standard deviation is higher than in **Figure 16** at each point; x-axis = numbering of signals of one block (sample index, n); y-axis = signals' heights (signal amplitude, a.u.).

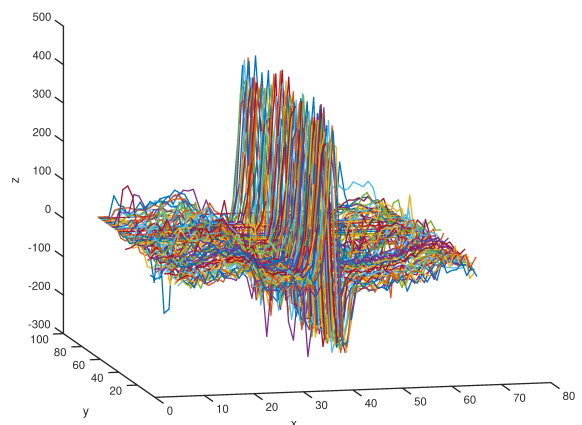


Figure 18 • A non-coherent block that consists of 100 cycles; x-axis = signal offset position with respect to the R-spike (sample index relative to R-spike, n); y-axis = depth to differentiate cycles aligned by R-spike (cycle index, unitless); z-axis = corresponding signal height (signal amplitude, a.u.).

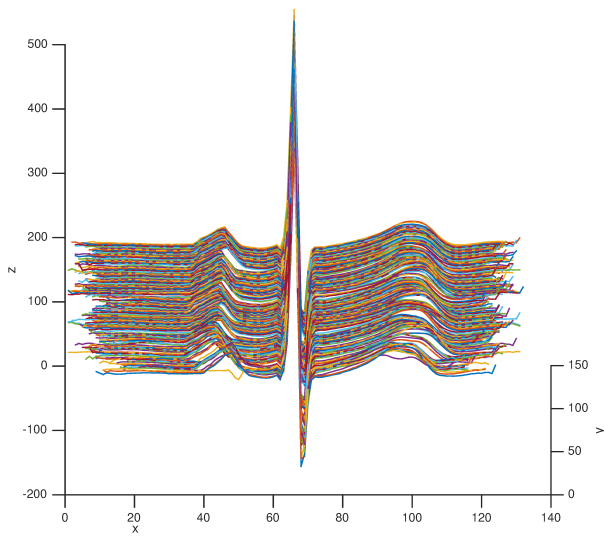


Figure 19 • Most faithful patterns of MIT-16420 arranged in chronological order: 150 patterns of 100 cycle blocks; x-axis = signal offset position with respect to the R-spike (sample index relative to R-spike, n); y-axis = depth to differentiate cycles aligned by R-spike (cycle index, unitless); z-axis = corresponding signal height (signal amplitude, a.u.).

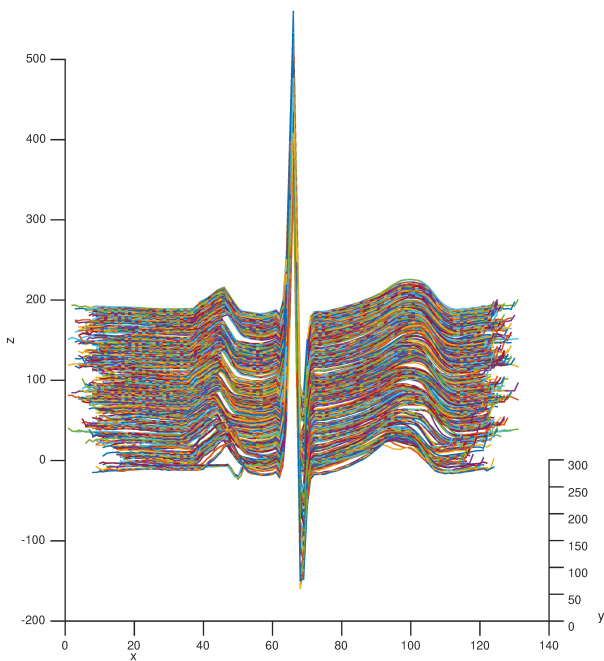


Figure 20 • Most faithful patterns of MIT-16420 arranged in chronological order: 300 patterns of 50 cycle blocks; x-axis = signal offset position with respect to the R-spike (sample index relative to R-spike, n); y-axis = depth to differentiate cycles aligned by R-spike (cycle index, unitless); z-axis = corresponding signal height (signal amplitude, a.u.).

4. Conclusions

It is possible to extract patterns automatically from massive noisy ECGs. The noise can be random or periodic of various frequencies at multifold spots within the recording. Moreover the recordings might contain interruptions due to electrode displacement and might be skewed by a wandering isoline due to motion.

Using quadratic variability, it is possible to flatten the wandering baseline, preserving the morphology of the cycles.

The monotone function that arises from peak counting allows for the identification of the R-spikes and the precise definition of the heart cycles. This allows for the automatic detection of the sections of the ECG lead that are too noisy or are otherwise devoid of clinically useful information. After removing these sections we obtain the good zones, which are subdivided into blocks. The latter are classified as coherent or non-coherent. The averages of the coherent blocks are the emerging patterns that have been identified and their chronological display offers a view of the dynamics of long stretches of the ECG.

Our focus tends to point to personalized medicine, towards prevention through early detection of suspicious deviations of normal electric behavior of the heart. All computations are in Matlab™.

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Author contributions

Conceptualization, M.P. and M.L.; methodology, M.P. and M.L.; software, M.P. and M.L.; validation, M.P. and M.L.; formal analysis, M.P. and M.L.; investigation, M.P. and M.L.; resources, M.P. and M.L.; data curation, M.P. and M.L.; writing—original draft preparation, M.P. and M.L.; writing—review and editing, M.P. and M.L.; visualization, M.P. and M.L.; supervision, M.P. and M.L.; project administration, M.P. and M.L.; funding acquisition, M.P. and M.L. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare that they have no competing interests.

Data availability statement

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